

DETAILED PROGRAM

MONDAY 21:

Introduction to Legged Robots (Michele Focchi) (5h)

- A1. Introduction to floating based robots
 - A2. Modeling of legged robots: dynamics of floating base robots (excess of coordinates/minimal coordinates modeling)
 - A3. Building a simulator: contact Models (rigid contact, compliant contacts), modeling of impact dynamics. Constrained dynamics obtained by contact null-space projection.
 - A4. From floating base dynamics toward Newton-Euler equations
 - A5. Basic Concepts in Legged Locomotion
 - A6. Quasi-static Trunk Controller
 - A7. Reduced models: Linear Inverted pendulum (LIP), Centroidal Dynamics, Single Rigid Body model, point mass model, inverted pendulum, spring-loaded inverted pendulum (SLIP)
 - A8. Balance metrics: Capture point
 - A9. Locomotion Frameworks
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TUESDAY 22:

Primer on optimization/optimal control (morning)(Marco Frego) (3h)

- B0. introduction to numerical optimization: LP, QP, SOCP (convex), non linear
- B1. Taxonomy explain direct/indirect method. Overview on Optimal control Problems transcription: Single shooting / Multiple Shooting/ Direct collocation (piece-wise knot, ZoH) for CoM trajectory optimization, examples single shooting, direct collocation.
- B2. Automatic differentiation/finite differences/mention analytical derivatives of Carpentier for rigid body dynamics
- B3. Types of Solvers: SQP(example), Interior Point, Active set, introduce sparsity concept
- B4. Discretization of dynamics (forward Euler, exact discretization with matrix exponentials (for linear systems), RK4

Advanced robot control (afternoon) (Andrea Del Prete)(3h)

- C1. Constraint inverse dynamics for floating base robots.
 - C2. Whole-body control: Quasi-static approach (pseudo-inverses) to control the Base or the Com
 - C3. QP-based controller (with inequality constraints): joint space/task space inverse dynamics (Torque limits, acceleration limits, velocity limits, position limits)
 - C4. QP-based controller in contact: friction constraints
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WEDNESDAY 23:

Trajectory Optimization (morning) (Enrico Mingo) (3h)

- E1. Introduction to Trajectory optimization for legged robots
- E2. Reference Generator
- E3. Gait scheduler: assembly of the support polygon constraints
- E4. Examples of linear optimization: with LIP (time invariant), solution of the trajectory optimization with LIP

- E5. Examples of linear optimization: with Single rigid body (time varying) Jared De Carlo, solution of the trajectory optimization with Single rigid body
- E6. Examples of non-linear optimization: SQP with Centroidal Model (Single Rigid BODY)
- E7. Concept of sparsity emerging in legged robot traj. optimization (band structure hessian), compare single shooting, direct collocation, multiple shooting. Examples.
- E8. Model Predictive Control

Sampling-based Control (afternoon) (Giulio Turrisi)(3h)

- F1. Sample-based approaches for ocp solution
 - F2. Random Sampling
 - F3. Splines, Warm Start, CPU-GPU
 - F4. CEM
 - F5. MPPI
 - F6. Application on legged robots
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THURSDAY 24:

State Estimation for legged robot (morning) (Marco Camurri) (3h)

- H1. Brief introduction to KF and EKF
- H2. Design of EKF for state estimation, fusing IMU, kinematics and laser scan

Reinforcement Learning (Afternoon) (Giulio Turrisi) (3h)

- G1. Introduction to Reinforcement Learning:
- G2. Value Iteration
- G3. Q-Iteration
- G4. Q-Learning
- G5. Deep Q-Learning
- G6. Policy Gradient
- G7. PPO

Social Dinner + delivery of the diplomas

FRIDAY 25:

LABS (Friday Morning)(3h)

Locomotion lab in **simulation**

Reinforcement Learning (Friday afternoon) (Giulio Turrisi) (3h)

- H1. Simulators in RL
- H2. Rewards
- H3. Observations
- H4. IMU vs Linear velocity vs Concurrent SE
- H4. Sim2Real
- H5. Asymmetric PPO

SATURDAY 26:

Optional tutored practical session with **Real Robots** Aliengo/go2

SUNDAY 27:

Trekking in Dolomites